FEATURE-BASED PROJECT DOCUMENT

SkillLens

Feature-Based Project Document

Project	SkillLens — Skill-to-Employment Intelligence Platform
Institute	Swami Vivekanand College of Engineering, Indore
Mentor	Ms. Amisha Jain
Authors	Teena Devray · Rohit Solanki · Vishal Panwar
Date	March 2026

1. Purpose

This document describes every feature of SkillLens from the user's perspective — what each feature does, what problem it solves, what the user sees and experiences, what inputs it requires, what outputs it produces, and what success looks like. It complements the SRS (which is requirement-centric) and the Module Document (which is code-centric) by being entirely user-centric.

Feature 1 — User Registration and Login

What problem does it solve?

Students need a persistent account so that their depth scores, interview history, roadmap, and readiness score are saved and improve over multiple sessions — not lost every time they close the browser.

What the user experiences

A clean two-page flow: Register (name, email, password, target job role) and Login (email, password). After login, the user lands on their dashboard. A JWT token is stored in the browser and silently attached to every subsequent API request — the user never sees or manages it manually.

Feature details

Detail	Specification
Input	Name, email, password (min 6 chars), target role (dropdown)
Output	JWT token; redirect to dashboard
Validation	Email format check; duplicate email rejection; password length enforcement
Security	Password hashed with bcrypt (10 rounds) before storage; JWT expires in 7 days
Error states	Duplicate email → 'An account with this email already exists'; Wrong password → 'Invalid credentials'
Success state	User lands on dashboard with their name displayed in the navbar

---

Feature 2 — Smart Resume Analyzer

What problem does it solve?

Students don't know which skills their resume communicates — or fails to communicate. This feature makes the invisible visible: it shows exactly which skills were detected and which are missing for their target role.

What the user experiences

The user drags and drops (or selects) their PDF resume. A progress bar appears. Within seconds, a results panel shows two columns: 'Skills detected in your resume' and 'Skills missing for your target role'. Each detected skill shows whether a depth quiz has been taken, and its current depth score if so. Missing skills show a prompt: 'Add this skill to your profile by taking a short quiz.'

Feature details

Detail	Specification
Input	PDF resume file — max 5 MB — text-extractable only

Processing	pdf-parse extracts raw text; case-insensitive string matching against 30-skill master list
Output	List of detected skills; list of missing skills for the user's target role
Edge cases	Scanned PDF → error message 'We couldn't extract text from this PDF. Please use a text-based PDF.'; Empty file → validation error before upload
What it does NOT do	Does not store the resume file; does not score the resume on formatting or length; does not generate a new resume
Success state	User can clearly see which skills to add or improve; each skill card links directly to its quiz

Feature 3 — Skill Depth Quiz

What problem does it solve?

Writing a skill on a resume is not the same as knowing it. This feature is the most original in SkillLens: it measures how deeply a student actually knows a skill — producing a percentage depth score — not merely whether they've written it on a document.

What the user experiences

The user selects a skill (or is prompted after resume analysis). A 10-question MCQ quiz begins — one question at a time, with four options. Questions are randomised on each attempt. After submission, the user sees their score, depth level, and a comparison to their previous attempt if they've retaken. A bar immediately shows how their depth compares to what the target company requires for that skill.

Feature details

Detail	Specification
Skills covered	React, Node.js, MongoDB, SQL, Python, Java, DSA, REST APIs, System Design, Git
Questions per skill	10 MCQs, randomised order per attempt
Depth levels	Beginner: 0–40% | Intermediate: 41–70% | Advanced: 71–100%
Scoring	One point per correct answer; depth score = (correct / 10) × 100
Retakes	Unlimited; each retake updates the stored depth score
Side effects	Depth score updates company match percentages for all 25 companies; Readiness Score recalculated
Success state	User knows their exact depth in a skill and how far they are from what a target company requires

Feature 4 — Company Match Engine

What problem does it solve?

Students apply randomly, without knowing which companies they are realistically suited for right now and which require specific improvement. This feature replaces random application with data-driven targeting.

What the user experiences

A ranked list of 25 Indian companies — from highest match to lowest — each showing a match percentage and a colour-coded readiness bar (green above 70%, amber 40–70%, red below

40%). Clicking any company opens a detail view with a gap analysis: which specific skills are pulling the score down, by how much, and the single most impactful action to take first.

Feature details

Detail	Specification
Company coverage	25 Indian companies: TCS, Infosys, Wipro, HCL, Cognizant, Tech Mahindra, Capgemini, Accenture India, Hexaware, Mphasis, Flipkart, Zomato, Swiggy, Razorpay, PhonePe, Meesho, Paytm, CRED, Juspay, Freshworks, Browserstack, Postman, Zoho, Persistent Systems, Mindtree
Match calculation	Weighted sum of min(studentDepth / requiredDepth, 1.0) × skillWeight for all required skills
Gap analysis	Top 3 skills by score impact, sorted by how much each is costing the match percentage
Verdict	One plain-language sentence: the single action that would most improve the match score
Update trigger	Match percentages recalculate automatically after every quiz submission or interview completion
Success state	User knows exactly which companies they can target now, which require 2–4 weeks of work, and which are longer-term goals

Feature 5 — AI Mock Interview

What problem does it solve?

Students practice interview answers in a vacuum — no feedback on whether their answer was technically correct, whether they communicated it clearly, or what the ideal answer would have looked like. This feature provides structured, specific, dimension-split feedback on every answer.

What the user experiences

The user selects their target role. Five questions appear one at a time. For each question, the user types their answer and clicks Submit. Within seconds, a feedback panel appears with six clearly labelled sections: Technical Score (n/10), Communication Score (n/10), What was technically correct or wrong, Was the answer clear and well-structured, The key concept that was missed, and a Model answer (two sentences) they can learn from. Missed technical keywords are shown as highlighted tags.

Feature details

Detail	Specification
Roles covered	Frontend, Backend, Full Stack, Data Analyst, Java Developer
Questions per session	5 questions, role-specific, from a static curated question bank
AI model	OpenAI gpt-3.5-turbo; prompt engineered to return strict JSON — never free text
Scoring dimensions	Technical accuracy (1–10) and communication quality (1–10) — scored independently
Feedback fields	Technical feedback, communication feedback, missed concept, model answer, missed keywords array
Keyword feedback loop	Missed keywords are written back to the user's skill profile to flag weak areas
Graceful degradation	If OpenAI API is unavailable, answer is stored without AI scoring; user is notified
Success state	User leaves each session with specific knowledge of what to study and how to structure their answers better

Feature 6 — Placement Roadmap

What problem does it solve?

Students know they need to improve but don't know in what order, on what schedule, or using which specific resources. Generic 'learn X' advice is useless. This feature produces a clickable, week-by-week plan that tells them exactly what to do, watch, read, and build — driven by their specific gaps for their specific target company.

What the user experiences

The user selects a target company and enters how many weeks they have before placement season. A roadmap generates within seconds. Each week is a card showing: the skill to focus on, the depth level to reach, the reason it was prioritised (e.g. 'Highest impact on your Razorpay score'), a YouTube video link, an article link, a practice platform link, and a concrete task. The user marks weeks complete as they progress.

Feature details

Detail	Specification
Inputs	Target company (from 25 options); weeks available (2–20)
AI role	OpenAI determines the ORDER of skills to address based on gap analysis — AI does not generate the resource links
Resources	Filled from a static curated JSON library — one video, one article, one practice link, one task per skill per level
Completion tracking	Each week is marked complete by the student; completion percentage feeds into Readiness Score (weight: 20%)
Regeneration	Roadmap regenerates automatically if the user changes their target company or if depth scores change significantly after quiz retakes
Success state	Student has a concrete, specific daily-action plan — not generic advice — tailored to their exact starting point and target company

---

Feature 7 — Progress Dashboard and Readiness Score

What problem does it solve?

Students prepare in isolation without knowing if they are making real, measurable progress. The dashboard answers the question every student has but no tool currently answers: 'Am I actually getting better, and am I ready?'

What the user experiences

A single-page overview showing: the Readiness Score (large, prominent, 0–100 with a label), a line chart of score over time, a horizontal bar chart of depth scores per skill, the top 3 companies they are closest to being ready for, their average technical and communication interview scores, roadmap completion percentage, and a 'Next Step' prompt — the one action that will most improve their score.

Feature details

Detail	Specification
Readiness Score formula	(avgDepthScore × 0.35) + (avgInterviewScore × 0.30) + (roadmapCompletion × 0.20) + (bestCompanyMatch × 0.15)
Score labels	0–39: Just starting | 40–59: Building up | 60–79: Almost there | 80–100: Placement ready
Charts	Line chart (score over time) and horizontal bar chart (skill depths) — both built with Recharts

Update triggers	Score recalculates after every quiz, every interview session, and every roadmap week completion
Next Step logic	Whichever of the four score components is lowest and most improvable drives the next-step prompt
Success state	Student can see their progress objectively, identify their weakest dimension at a glance, and know exactly what to do next

8. Feature Summary Table

Feature	User Value	Technical Complexity	Uniqueness
User Auth	Account persistence across sessions	Low	Standard
Resume Analyzer	Know which skills your resume communicates	Low–Medium	Standard
Skill Depth Quiz	Know HOW WELL you know each skill	Medium	HIGH — no competitor does this
Company Match Engine	Know which companies you're ready for RIGHT NOW	Medium	HIGH — weighted + depth-aware
AI Mock Interview	Get specific, structured feedback per dimension	Medium–High	HIGH — dual scoring + keyword loop
Placement Roadmap	Get a clickable, curated weekly action plan	Medium	HIGH — resource-linked, gap-driven
Dashboard + Readiness Score	Know if you're actually improving	Medium	HIGH — synthesises all modules